Samarth Gupta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.60
Guru Harkrishan Public School, Jammu	Class XII (The Jammu & Kashmir Board of School Education)	2024	94.20%
Delhi Public School, Jammu	Class X (Central Board of Secondary Education)	2022	90.80%

WORK EXPERIENCE

Founder's Office and Marketing Intern | PicaPool

- Assisted the Founder's Office in planning and executing student marketing campaigns across multiple college campuses.
- Conducted competitor research and analyzed student engagement data to identify content ideas and campaign opportunities.
- Coordinated with campus ambassadors to support promotional activities, event registrations, and feedback collection.
- Tracked campaign performance using basic metrics such as CTR, sign-ups, and social media engagement, and shared weekly insights with the team.

PROJECTS

Student Organization Finance Manager

- Built a **standalone Progressive Web App (PWA)** to manage budgets, reimbursements, and event planning for **15 active student clubs**, replacing scattered spreadsheets with a centralized, real-time administrative dashboard.
- Programmed core logic with **vanilla JavaScript and localStorage** to process multi-step workflows, integrating **offline Service Workers** and a custom **PIN lock** to secure sensitive club financial data entirely on the device.

Hyderabad Transit Planner

- Built a comprehensive **full-stack web application** mapping over **1,500+ TSRTC city bus routes** and **4,700+ bus stops**, directly addressing public transit navigation gaps and route disconnects **across the entire city of Hyderabad**.
- Integrated localized campus transport data, including **IIT Hyderabad shuttle schedules** and regional shared auto connections, enabling students to seamlessly plan, customize, and execute multi-leg travel from campus to **any location.**
- Engineered a custom multi-criteria **scoring algorithm** to evaluate and **rank optimal routes** based on total travel time, required transfers, and walking distance, **presenting alternative paths** on an **interactive map interface.**

Extended Physics-Informed Neural Networks (XPINNs) for Hyperbolic PDEs

- Solved a **coupled hyperbolic PDE system** with discontinuous initial conditions by decomposing the spatial domain into interface-coupled **6-layer MLP sub-networks** to capture sharp shock transitions across both subdomains.
- Engineered a **physics-based loss function** via **batched Adam gradient descent**, validating predictions against **exact analytical benchmarks** to verify smooth solution joining at $x=0$ across multiple fixed spacetime intervals.

POSITIONS OF RESPONSIBILITIES

Cultural & Literary Secretary | Student Gymkhana, IIT Hyderabad

- Lead the Cultural & Literary Council, directing **13 active clubs and 250+ student members** to plan, market, and

execute **over 70+ diverse annual** institute events, driving active participation for **6,000+ students across campus.**
- Represent the Cultural Council before the institute administration, coordinating **budget approvals**, event clearances, overall stakeholder communication, and policy implementation across all student clubs for institute-wide activities.
- Formulate operational policies, allocate resources, and **build dedicated teams**, acting as the lead institute cultural representative to direct coordination with hosting IITs, home leadership, and contingents for **Inter-IIT Cultural Meet**.

Head | Cultural Council, IIT Hyderabad

- **Headed 4 core verticals** -- Events, PR, Logistics & Operations, and Design -- as Head of the Cultural Council.
- Managed allocation of **15L+ budget** across cultural initiatives, coordinating resource planning and financial operations.
- Conceptualized and executed logistics and operations for **10+ institute events and Inter-IIT preparations.**
- Grew the council's Instagram presence, increasing impressions and follower count by **40%+** through content strategy.

Public Relations Coordinator | Elan & nVision, IIT Hyderabad

- Spearheaded PR and media outreach for flagship fest Elan & nVision, driving sponsor relations for **30K+ attendees**.
- Managed **10K+ follower Instagram account**, directing reel strategy and sponsor campaigns to drive **1M+ views**.

Interview & Hosting Coordinator | Media Council, IIT Hyderabad

- Hosted **10+ formal and informal institute events**, moderating stage proceedings, interviews, and podcasts.
- Conceptualized, scripted, and produced **Class Songs** for graduating batches, garnering **80K+ cumulative views.**

Events & Public Relations Coordinator | SPIC MACAY, IIT Hyderabad

- Coordinated PR and operations across 8 events, **hosting 2 sessions** and managing stage flow for key live events.

SKILLS

Programming Languages & Databases: C, Python, SQL, HTML, CSS

GenAI & AI Tools: ChatGPT, Claude, Gemini, Perplexity, NotebookLM, ElevenLabs, HeyGen, Runway, Antigravity

Libraries & Frameworks: NumPy, Pandas, Pytorch, SciPy, Matplotlib, React, FastAPI, Seaborn, Scikit-learn, Plotly

Developer Tools: VS Code, PyCharm, Jupyter Notebook, Fusion 360, Google Colab, MySQL, Canva, Antigravity, GitHub

EXTRACURRICULAR

- Represented IIT Hyderabad at the Inter-IIT Cultural Meet 8.0, competing in high-stakes theatre and comedic arts.
- Hosted cultural events while winning Inter-Department medals, securing points toward the Overall Championship.
- Active hybrid athlete combining strength training with long-distance running, training to complete the Procam Slam.